Abhishek Yadav | IIT Madras

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EDUCATION

PhD – Petroleum Engineering	Indian Institute of Technology Madras	9.88	2024 - Present
M. Tech – Petroleum Engineering	Indian Institute of Technology Madras	9.61	2023 – 2024*
B. Tech - Petroleum Engineering	Rajiv Gandhi Institute of Petroleum Technology	8.15	2018 - 2022
XII (RBSE)	Mehta Public Sr. Sec. School, Jaipur	87.20 %	2017
X (RBSE)	Mehta Public Sr. Sec. School, Jaipur	88.00 %	2015

*Upgraded to PhD

SCHOLASTIC ACHIEVEMENTS

- Winner of **SPE South Asia Student Paper Contest**, 2026 in PhD division for research in depleted oil geothermal systems.
- Selected to represent South Asia region at the International Student Paper Contest, **2026 SPE ATCE**, Houston.
- Best Poster Presentation Award at **SCORE 2025**, IIT Madras.
- Achieved All India Rank 76 in **GATE 2024** and 108 in **GATE 2023**.

PUBLICATIONS

- **A. Yadav**, M.I. Ansari, S.K. Govindarajan, "Multi objective optimization of a discrete fracture geothermal reservoir using Bi-LSTM network," Energy, Vol. 347, 140320, 2026. (**Published**) **[Link]**
- **A. Yadav**, S.K. Govindarajan, "Numerical investigation of energy extraction from a depleted oil geothermal reservoir," SPE Annual Technical Conference and Exhibition (26ATCE), 2026. (**Published**) **[Link]**
- **A. Yadav**, M.I. Ansari, S.K. Govindarajan, "Fractured Geothermal Reservoir Performance Estimation Using Supercritical CO2," Journal of Thermal Science and Engineering Applications, 18(4), 041011, 2026. (**Published**) **[Link]**
- S.K. Govindarajan, M.I. Ansari, T.N.V. Pavan, **A. Yadav**, P. Chaudhary, "Limitations Associated with Extended Darcy's Law as Applied in the Petroleum Industry," Improved Oil and Gas Recovery, 9, 2025. (**Published**) **[Link]**
- S.K. Govindarajan, M.I. Ansari, T.N.V. Pavan, **A. Yadav**, P. Chaudhary, "Where Extended Darcy Models Lose Physical Fidelity in Multiphase Petroleum Flow," Science Proceedings, 12, 2025. (**Published**) **[Link]**
- M.I. Ansari, **A. Yadav**, P. Chaudhary, A. Hanegaonkar, S.K. Govindarajan, "Numerically Evaluating the Performance and Sustainability of an Enhanced Geothermal Fractured Reservoir," 6th EAGE Global Energy Transition Conference & Exhibition (GET 2025), 2025. (**Published**) **[Link]**
- J. Nembhnani, **A. Yadav**, S. Biswas, "Developing an intelligent fire alarming, monitoring and rescuing system using UAV," The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences, 2020. (**Published**) **[Link]**
- **A. Yadav**, P. Chaudhary, S.K. Govindarajan, "Numerical investigation of a coupled THM model in a geothermal reservoir with a discrete fracture network," 7th EAGE Global Energy Transition Conference & Exhibition (GET 2026), 2026. (**Accepted**)
- P. Chaudhary, **A. Yadav**, S.K. Govindarajan, "Numerical evaluation of CO2 sequestration performance in a heterogeneous deep saline aquifer by hydro-mechanical coupling," 7th EAGE Global Energy Transition Conference & Exhibition (GET 2026), 2026. (**Accepted**)

PROFESSIONAL EXPERIENCE

Aitele Research (Intern)
May-June '24

AI-Powered Dynamic Dashboard Generator

- Developed a machine learning model trained on over **2000** dashboard datasets.
- Designed a streamlined **GUI** for user input, facilitating efficient data visualization.
- Built core **NLP** functionalities for understanding user descriptions and extracting key information

Torrent Gas Pvt. Ltd.
July '22- July '23

Graduate Engineer Trainee (GET), CNG-(O&M)

- Managed operations and maintenance for over **40 CNG** stations, leading a team of **80+** personnel.
- Contributed to an **11%** increase in equipment availability by optimized maintenance and reduced downtime.
- Coordinated with OMC authorities to ensure compliance and managed material inventory through **SAP**.

Cairn Oil & Gas, Vedanta Limited (Intern)
May - June '21

Intervention in Horizontal wells: Challenges and way Forward

- Analysed historic **well interventions** and addressed challenges in complex well completions.
- Suggested reliable alternatives for production logging and sand cleanout in **ABH horizontal wells**.

PROJECTS

Facies Classification with Machine Learning
- Kaggle

- Developed a machine learning model using well log data using **Logistic Regression**, **SVM**, and **KNN** algorithms.
- Achieved **F1 scores** of **0.72** for exact and **0.93** for adjacent facies classification with **KNN**.
- Deployed a **Streamlit UI** for intuitive user interaction and facies classification results.

M. Tech Project
IIT Madras
June '24 – Dec '24

Numerical Investigation of heat extraction from Geothermal Reservoir

- Conducted a **numerical investigation** on heat extraction from geothermal reservoirs using **COMSOL**.
- Compared the performance of **supercritical CO_2** and water as geofluids for heat extraction.
- Developed predictive models with **machine learning** to forecast production temperatures.

POSITIONS OF RESPONSIBILITY	
Saathi Club	• Mentoring and guiding newly admitted students in academic matters and supporting their transition to the academic environment at IIT Madras
Secretary	• Society of Petroleum Engineers, IIT Madras Student Chapter
Teaching Assistant	• **Courses**: PE6030 Reservoir Engineering, PE6312 Enhanced Oil Recovery, PE6031 Reservoir Simulation IIT Madras
Joint Secretary	• Science and Technology Council, RGIPT

SKILLS	
Prog. Language	• Python, SQL, Frameworks: Pandas, Numpy, Matplotlib, Scikit-Learn, Pytorch, NLTK, PorePy
Software	• MS Office, Power Bi, MySQL, SAP, COMSOL, CMG

COURSES		
Finite Element Method: Concepts and Applications in Geomechanics	Numerical Solutions of Partial Differential Equations	Advanced Seismic data analysis and Interpretation
Reservoir Simulation	Enhanced Oil Recovery	Petroleum Geomechanics
Exploration and Formation evaluation	Surface Facilities	Artificial Lift Technologies
Data Science: Theory and practice	Machine Learning Specialization (Coursera)	Python Project for Data Science (Coursera)
Machine Learning with Python	Data Analysis with Python (Cognitive Class)	Drilling Technology

EXTRA-CURRICULAR ACTIVITIES	
Volunteer	• Worked with Samagra Foundation to find jobs for individuals with limited educational qualifications.
Sports	• Badminton, Skating
Languages	• English, Hindi

RBSE - Rajasthan Board of Secondary Education, GATE - Graduate Aptitude Test in Engineering, CNG – Compressed Natural Gas,
ABH - Aishwarya Barmer Hill, OMC - Oil Marketing Company